

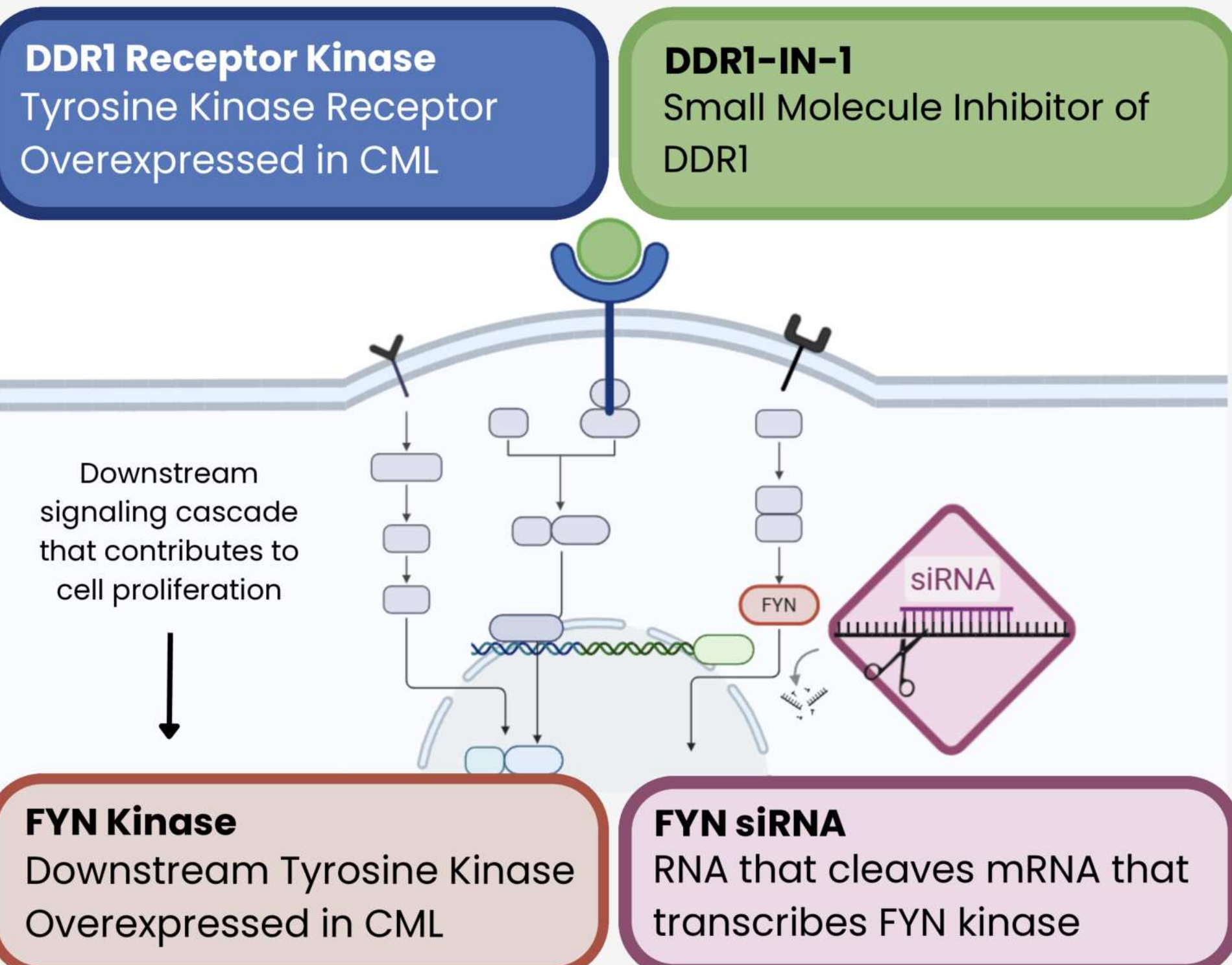
Assessing The Impact of The Dual-inhibition of DDR1 and FYN on Cell Viability in Human Chronic Myelogenous Leukemia

Background

Chronic Mylogenous Leukemia(CML)

- CML is a blood cancer driven by **overactive tyrosine kinase** signaling
- TKIs improve outcomes, but **resistance and relapse** occur due to activation of alternative survival pathways

Proposed Solution: Dual Kinase Inhibition



Graphic created by researchers using Biorender, 2026

Hypothesis: Dual Inhibition is More Effective

If:

CML cells are treated with DDR1-IN-1 + FYN siRNA

Then:

Cell viability will decrease more than with single treatments

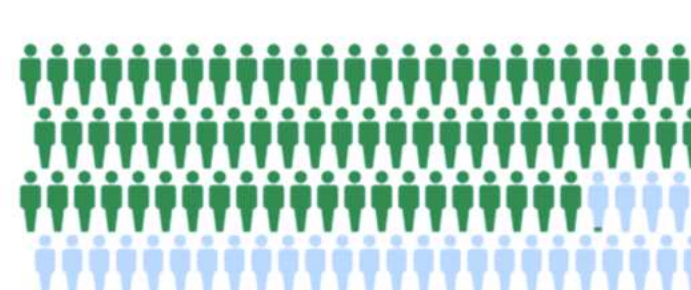
Justification:

DDR1 and FYN activate complementary survival pathways, so dual inhibition **blocks compensation** and **increases cell death**. Targeting both DDR1 and FYN may enhance therapeutic efficacy by **disrupting parallel kinase signaling pathways** that contribute to proliferation, survival, and treatment resistance in CML

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Significance

28.9% Relative 5-Year Death Rate



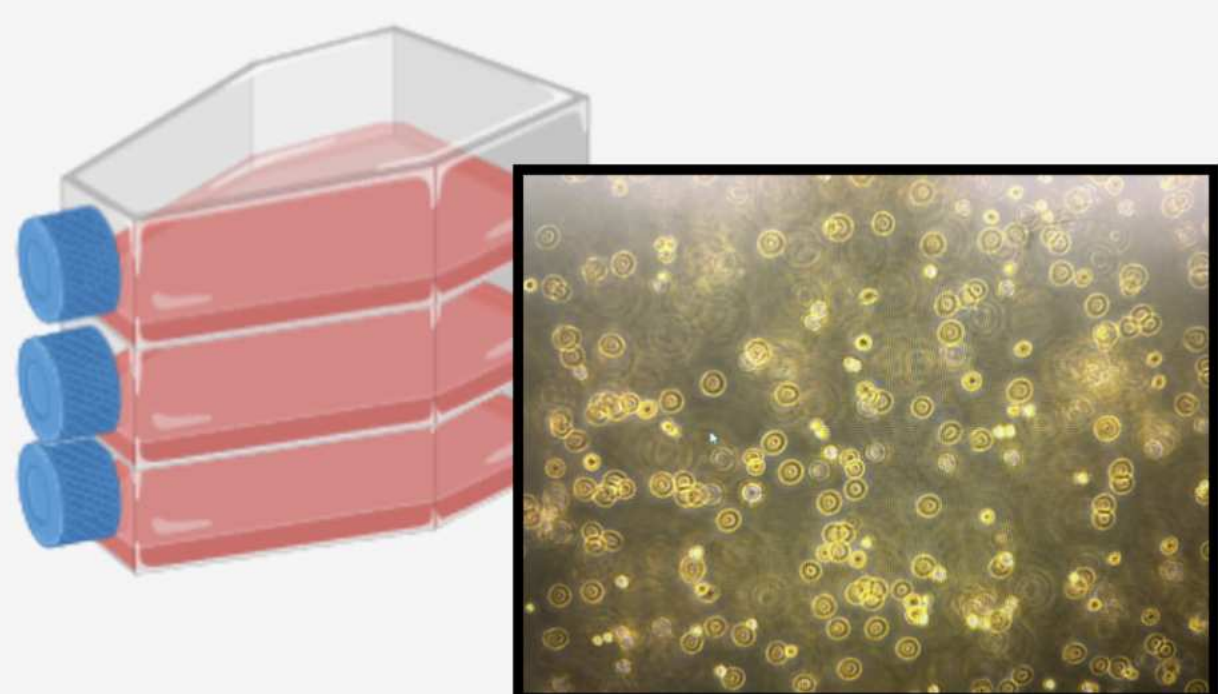
Green figures represent those who have survived more than 5 years while blue figures represent those who have died from CML.

(Chronic Myeloid Leukemia - Cancer Stat Facts, n.d.)

- Current therapies fail long-term due to **drug resistance**
- Cause: **alternative survival pathways** which may be overcome by DDR1 + FYN inhibition
- Advances **combination therapy** strategies

Methods

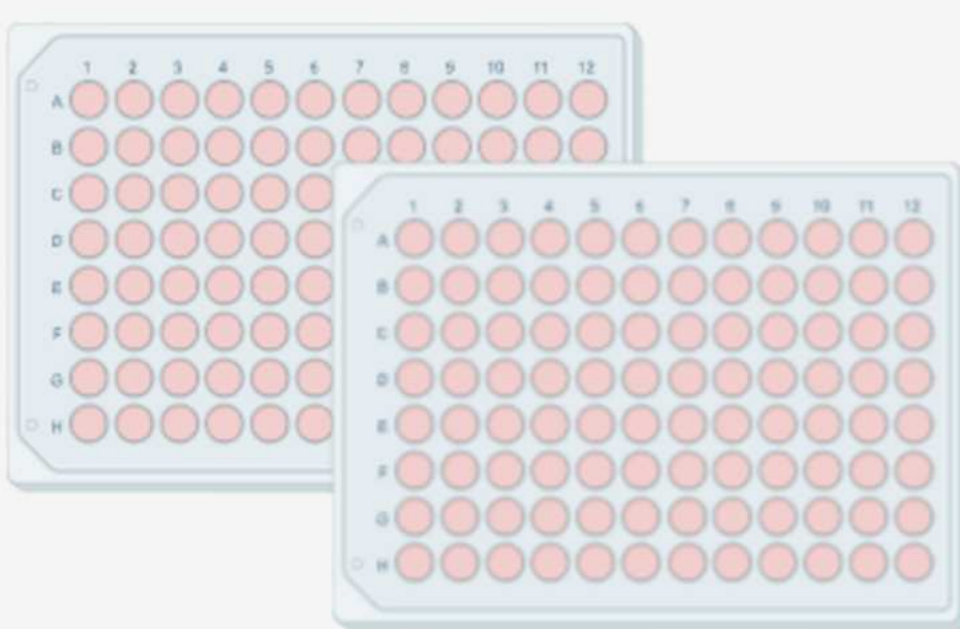
1) Cell Culture



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Figure. K562 Human Chronic Myelogenous Leukemia cells at 80% Confluency

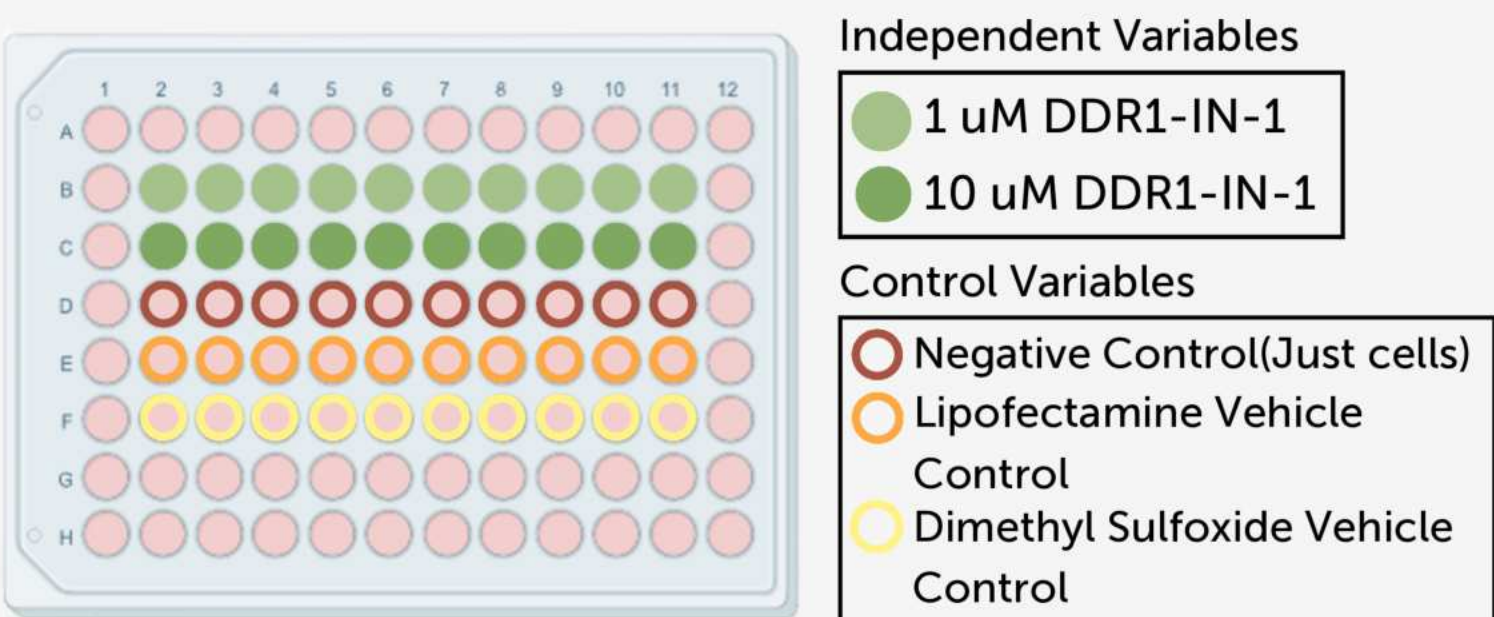
2) Seed K562 Cells



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K562 Cells were seeded at a desired confluency in 2 96-well-plates

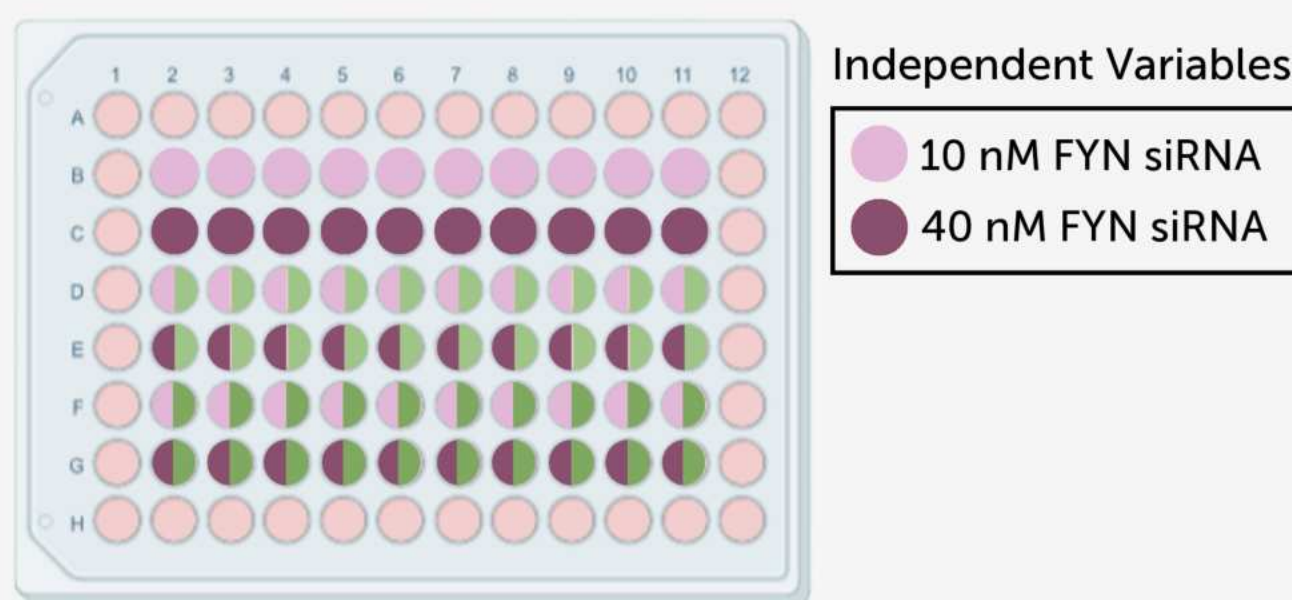
3) Administer DDR1-IN-1



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DDR1-IN-1 was administered using DMSO as a vehicle

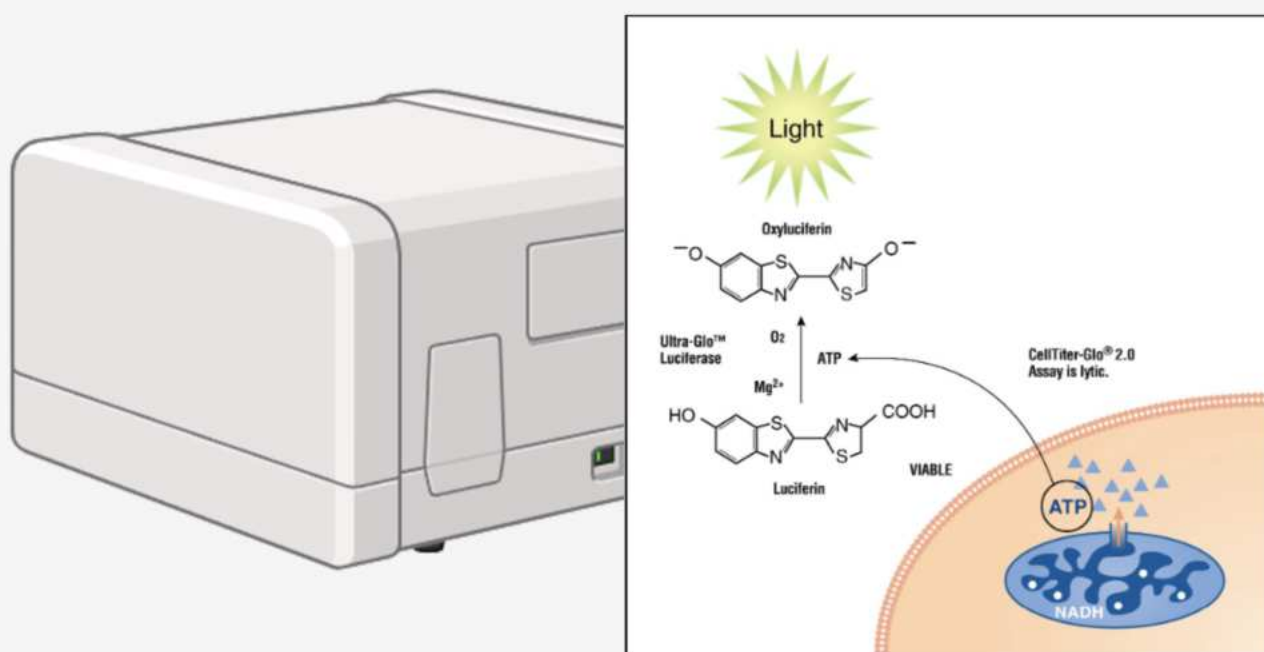
4) Transfect FYN siRNA



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FYN siRNA was administered using Lipofectamine and OptiMEM as a vehicle

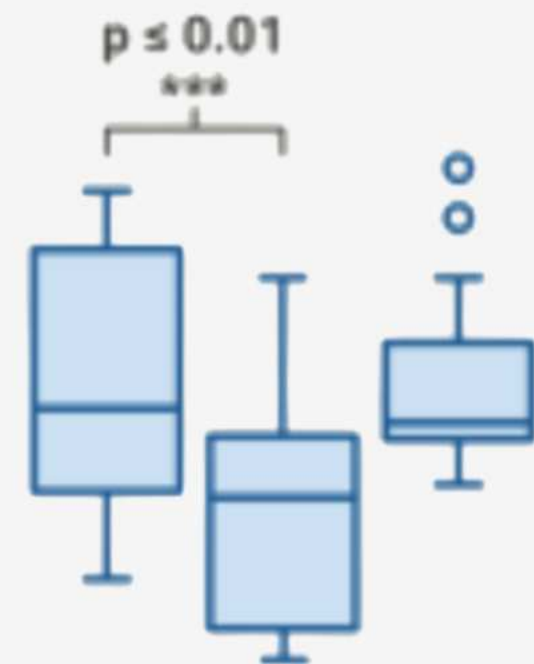
5) Cell-Titer Glo Assay



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CellTiter-Glo reagent is added to cells, where it reacts with metabolically active cells to produce a measurable signal **proportional to cell viability**.

6) Statistical Analysis



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A **Two-Way ANOVA** test is performed to determine whether treatments groups cause a significant difference in viability

Discussion

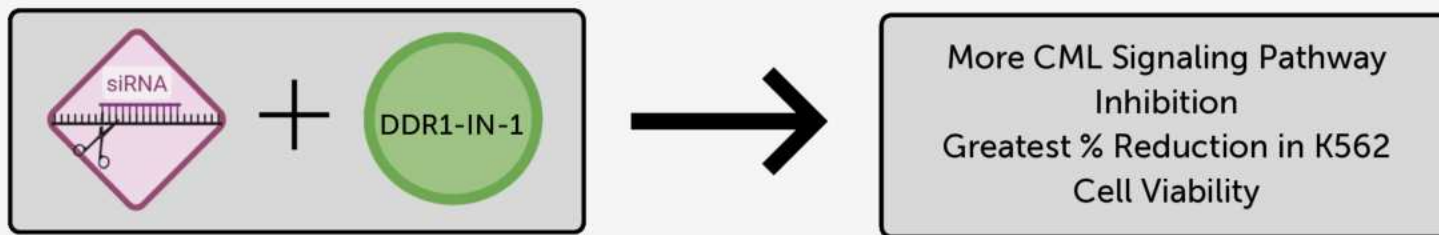
Collected Data **Supported Hypothesis:**
Dual inhibition had a significantly greater effect on reducing cell viability compared to single treatments.

Cell Viability (CellTiter-Glo Assay)

- Significant differences** observed between combination treatments and single/vehicle controls
 - Combination groups** showed the **greatest reduction in viability**
- Single treatments (DDR1-IN-1 or FYN siRNA) showed moderate or minimal effects
- Conclusion: Combination treatment most effectively reduced cell viability

Statistical Analysis (2-Way ANOVA)

- A **significant treatment effect** was observed ($p = 0.01$), indicating differences between groups are unlikely due to chance
- Post-hoc comparisons identified **significant differences** between combination treatments and both single treatments and controls
- The strongest significance was observed in **high-dose** combination conditions



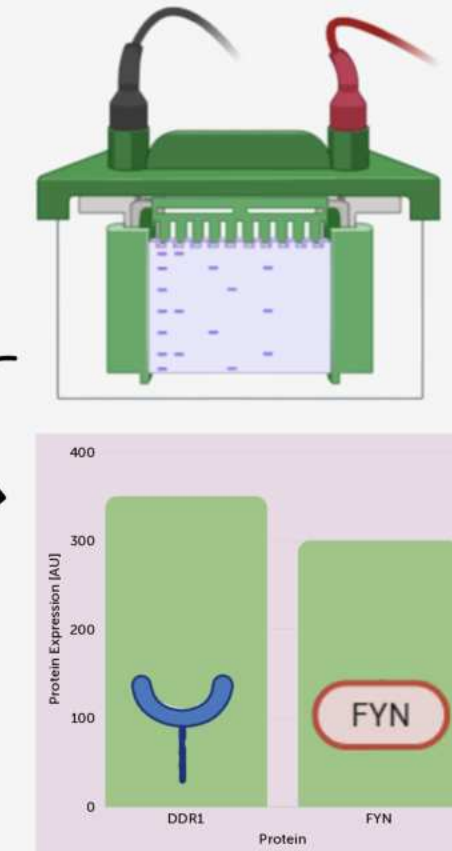
Implications

- Dual DDR1 and FYN inhibition** represents a **novel multi-target** therapeutic strategy for overcoming resistance in CML
 - May improve outcomes in patients with **TKI-resistant** disease
 - Supports the broader use of **combination-targeted therapies** in cancer treatment
- Findings contribute to understanding how **pathway redundancy** drives treatment resistance

Future Work

- Test additional** DDR1-IN-1 and FYN siRNA **concentration combinations** to determine the **optimal therapeutic** window
- Increase trial number and **improve transfection consistency** to reduce variability
- Conduct **mechanistic studies** (eg., protein expression of DDR1, FYN, and downstream markers)
- Validate findings in **other leukemia cell lines** or primary patient samples

Western Blot Assay



Graphic created by researchers using Canva, 2026

References

Chronic Myeloid leukemia - Cancer Stat Facts. (n.d.). SEER. <https://seer.cancer.gov/statfacts/html/cmyl.html>
Jabbour, E., & Kantarjian, H. (2025). Chronic myeloid leukemia. JAMA, 333(18), 1618. <https://doi.org/10.1001/jama.2025.0220>
Key Statistics for Chronic Myeloid Leukemia (CML). (2026, January 13). <https://www.cancer.org/cancer/types/chronic-myeloid-leukemia/about/statistics.html>
Li, S. (2008). Src-family kinases in the development and therapy of Philadelphia chromosome-positive chronic myeloid leukemia and acute lymphoblastic leukemia. Leukemia & Lymphoma/Leukemia and Lymphoma, 49(1), 16-26. <https://doi.org/10.1080/10428190701713689>